Explainability as a Diagnostic Tool for Multilingual Failure in Audio Deepfake Detection

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Abstract. A substantial degradation in performance of transformer based audio deepfake detectors is observed in cross lingual evaluations outside their training matched english data , what remains unclear are the mechanisms causing that performance decline. This paper uses post-hoc explainability as a diagnostic tool by studying a wav2vec 2.0 XLS-R + AASIST transformer architecture. This study constructs Hindi and English speech from Common Voice dataset with Griffin Lim copy synthesis and then conducts a controlled comparison between languages. The synthesis is done in a way that fixes the vocoder family across both languages, mirroring the Griffin Lim family of ASVspoof 2019 attack A11. Under identical conditions, it is observed that the EER rises from 16.18% on English to 30.49% on Hindi – a 14.31 point gap, 95% CI [11.38, 17.92]. After accounting for differences in acoustic support, the gap remains significant – 8.52 points, [4.56, 12.76]. In another experiment, despite trimming leading/trailing silence from every utterance the gap still persists – 12.80 points, [9.54, 16.12]. In case of fake speech, there is a shift in attribution from low frequencies (73.48% $\rightarrow$ 44.58%) to mid/high frequencies (19.82%/6.70% $\rightarrow$ 30.81%/24.61%), and paired real-fake controls show that this redistribution caused by reconstruction is significantly larger in Hindi. Broad temporal concentration and attribution due to speech/silence remain similar across both languages, thus evidencing a relocation within speech and away from silence, separating this failure from the English silence shortcut.

Keywords: explainable AI · audio deepfake detection · wav2vec 2.0 · cross lingual generalization · shortcut learning · Griffin Lim

Table 1: Index

1 Introduction

A near perfect performance is achieved by self supervised speech models, like wav2vec 2.0 XLS-R + AASIST graph attention based backend in audio deepfake detection on their training matched benchmark: the model checkpoint used here reaches an Equal Error Rate (EER) of 0.22% on the ASVspoof 2019 LA evaluation partition (Section 2.1).

In contrast, 37.71% EER on a multilingual development set and 42.6% EER on a multilingual evaluation set spanning English, Singapore English, Japanese, Vietnamese, Mandarin and Taiwanese Mandarin under realistic media formations against 1% EER in its english only condition, has been reported by the RADAR Challenge 2026 [9] as baseline which uses the wav2vec 2.0 XLS-R + AASIST checkpoint – identical to the one used in this paper.

This collapse has been established in existing cross lingual studies. However, using explainability to understand why it happens at the level of what the model is actually focusing on has not yet been established.

Two different modes of failure can be attributed to the same EER drop: (i) The detector using specific language or speaker dependent shortcuts like prosody, silence etc. that are not transferrable across languages. or (ii) The artifact features as a result of synthesis which the detector learnt in training might not be as separable, or may simply not exist.

Both of them are solved differently: applying shortcut mitigation techniques for (i), and changes in architecture and training in multiple languages for (ii).

The approach:

1. Analysis of attribution maps and faithfulness metrics on English, building on Grinberg et al. [3].
2. Replication of the known silence shortcut on English established by Müller et al. [5].
3. Identical griffin lim copy synthesis to construct fake Hindi and English speech from real CV data allowing us to use the same vocoder family as ASVspoof 2019 attack A11 [8].
4. Comparison of attribution analysis, faithfulness evaluation and explanation drift between the synthesized English and Hindi speech to study the cross lingual failure.

Author Contributions

Pranav Bansal: Conceptualization, methodology, investigation, experimental design, data curation, implementation of the research experiments, analysis and interpretation of results, and writing of the original manuscript.

Kush Paruthi: Software development and implementation of the dashboard for visualization and presentation of the research results.

Rajashree Sengupta: Software development and implementation of the web application serving as a demonstration interface for the project.

All authors reviewed and approved the final manuscript.

2 Research Methodology

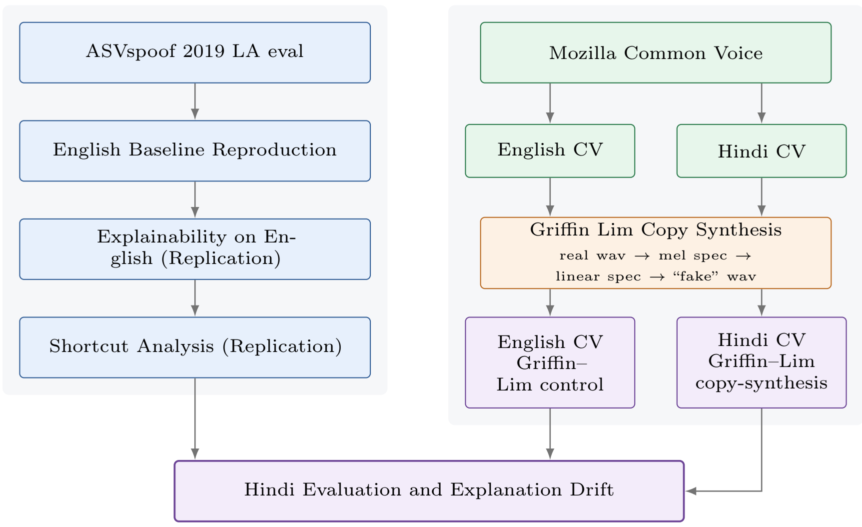

Fig. 1: Research Workflow

2.1 English Baseline Reproduction

The pretrained wav2vec 2.0 XLS-R + AASIST checkpoint is run on the ASVspoof 2019 LA eval partition (in domain) and EER is computed to be ≈0.2%.

Table 2: English baseline (ASVspoof 2019 LA eval, 71,237 utterances: 7,355 bonafide, 63,882 spoofed). These numbers were independently re-verified in the silence experiment (Section 2.3).

Partition	EER (%)	threshold
ASVspoof 2019 LA eval (in-domain)	0.22	2.00

2.2 Explainability on English (Replication)

Replication of attribution analysis following Grinberg et al. [3] is conducted on 100 samples (50 bonafide, 50 spoof) drawn from the 2019 LA eval partition. Occlusion (three window sizes), Integrated Gradients attributions, and faithfulness metrics are calculated at the three top K fractions for each method.

Two interesting patterns emerge:

(i) comprehensiveness grows with K (up to 4.3 logits for IG at 20%) and remains strongly positive for both methods which means that removing the highlighted regions reliably destroys the bonafide evidence.

(ii) sufficiency remains well below the original score even at K=20% (mean values stay negative, i.e. spoof leaning) meaning the highlighted regions alone are not enough to reconstruct the decision, which

Table 3: faithfulness on English (mean $\pm$ std over 100 samples), IG and 50 ms occlusion.

Method	Top K	Comprehensiveness	Sufficiency
IG	5%	2.40 ± 2.86	-4.13 ± 2.21
	10%	4.02 ± 3.89	-3.43 ± 3.13
	20%	4.32 ± 4.42	-1.58 ± 4.70
Occlusion (50ms)	5%	1.34 ± 1.77	-1.74 ± 2.14
	10%	2.41 ± 2.50	-2.67 ± 2.62
	20%	3.34 ± 3.05	-2.22 ± 3.54

strongly point towards the detector's evidence being temporally diffused. IG shows more comprehensiveness as compared to occlusion at every K, as established in [3].

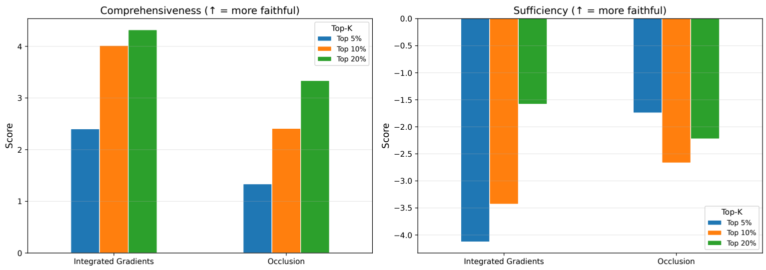

Fig. 2: Faithfulness (bonafide logit scale) for IG and occlusion at all three window sizes, $K \in \{5\%, 10\%, 20\%\}$.

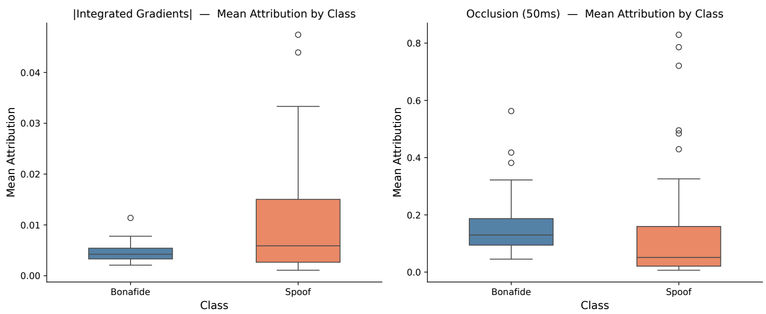

Fig. 3: Mean attribution over time for bonafide vs. spoof samples (IG and 50 ms occlusion).

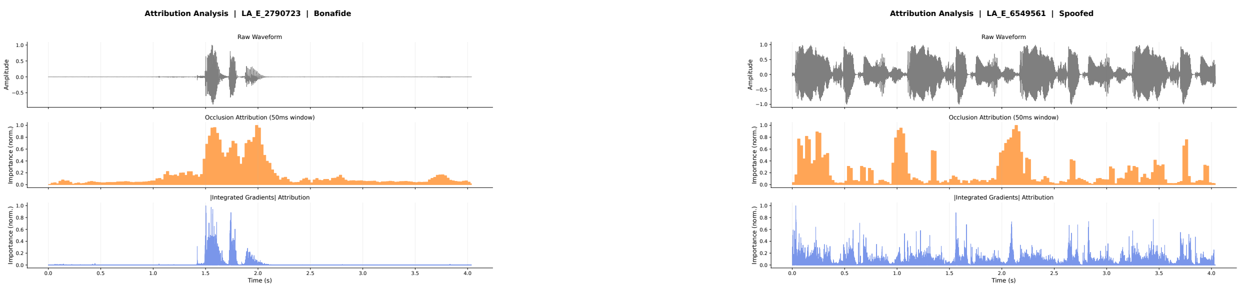

Fig. 4: Example attribution maps: waveform, 50 ms occlusion, and |IG| for one bonafide (`LA_E_2790723`, left) and one spoof (`LA_E_6549561`, right) evaluation utterance.

2.3 Shortcut Analysis (Replication)

Following Müller et al. [5], we test whether silence acts as a shortcut for this detector on ASVspoof 2019 LA eval, at a threshold of `top_db=40`.

Table 4: EER before/after silence trimming (full 71,237 trial eval set). The prediction flip rate reported at the original decision threshold is found to be highly asymmetric (60.2% of bonafide utterances flip to the spoof decision after trimming. In contrast, 0.37% of spoof utterances flip to bonafide).

Condition	EER (%)	Pred. flip rate (overall)
Original	0.22	—
Silence-trimmed	8.81	6.5%

EER degrades from 0.22% to 8.81% (a +8.6 percentage point increase) by removing only leading/trailing silence. This mechanism is observed by the flip asymmetry – long leading silence is a key supporting evidence for bonafide utterances, while spoof utterances marked by short leading silence remain unaffected in comparison.

This confirms that the silence shortcut documented by [5] for a different architecture, generalizes to SSL based detectors.

Table 5: Silence duration only shortcut probe (Comparing bonafide and spoof using only the silence duration as score. longer silence $\rightarrow$ bonafide). Both features separate the classes far above chance (Mann–Whitney $p < 10^{-3}$ for both).

Feature	EER (%)	Mean bona (ms)	Mean spoof (ms)
Leading silence	15.88	853	178
Trailing silence	38.12	400	185

The leading silence only probe reaches 15.88% EER which closely replicates the 15.1% EER documented by Müller et al. for their silence only probe [5], despite our detector being a different model class. Figure 5 shows the underlying distributions.

Table 6: Ratio of top 10% attribution falling in silence (relative to chance = silence fraction of the signal). Values below 1× chance mean the model attends to silence less than a uniform attribution over the waveform would.

Method	Class	Overlap frac.	× chance
IG	bonafide	0.017	0.05×
IG	spoof	0.023	0.31×
Occlusion (50ms)	bonafide	0.006	0.02×
Occlusion (50ms)	spoof	0.076	0.75×

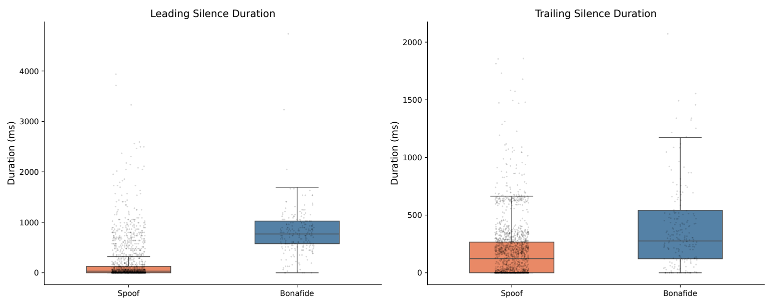

Fig. 5: Leading/trailing silence duration distributions by class on ASVspoof 2019 LA eval.

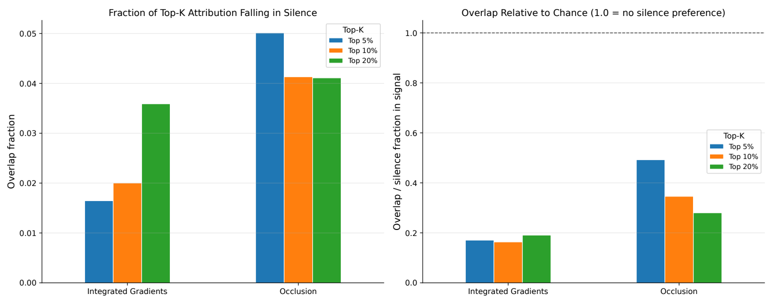

Fig. 6: Top K attribution overlap with detected silence, relative to chance.

A dissociation is exposed in Table 6 – silence duration is a key predictive element of the label (Table 5) and strongly drives the detector's decision (Table 4). Despite that the model's input attributions on silent regions fall below chance level meaning that the detector's evidence sits on speech instead of silence. This shows that a shortcut can drive decisions without dominating the attribution map.

2.4 Hindi Dataset Construction via Griffin Lim Copy Synthesis

Method Attack A11 in the ASVspoof 2019 LA taxonomy uses the Griffin Lim algorithm [12] for waveform generation [8,11]. The pipeline used here is:

$$\text{real wav} \xrightarrow{\text{Mel}} \text{mel spec} \xrightarrow{\text{InverseMelScale (SGD)}} \text{linear spec} \xrightarrow{\text{Griffin-Lim}} \text{“fake” wav}$$

implemented with `torchaudio.transforms.MelSpectrogram`, `InverseMelScale`, and `GriffinLim` ($n_{\text{fft}} = 1024$, hop$= 256$, $n_{\text{mels}} = 128$, 16 kHz, matching the ASVspoof 2019 LA convention). For Hindi fakes to stay a faithful reproduction of the attack family in our scope (A11), the Griffin-Lim iteration count is fixed at 32, matching A11's documented specification. SGD based pseudo-inversion via `InverseMelScale` serves as a standard, defensible choice for the mel to linear conversion step since A11's exact implementation cannot be independently verified from the public description.

The pipeline further incorporates `Dynamic peak fallback` to prevent high frequency distortion due to mathematical flat topping on excessively loud utterances. The waveform is gracefully peak normalized to 0.99 if matching the original source RMS causes digital clipping. Moreover, strict length safe processing guarantees that batch zero padding cannot alter amplitude targets or STFT outputs. Thus, pure Griffin Lim artifacts are kept isolated from unintentional digital noise.

Source Corpus and Quality Filtering Real Hindi speech is obtained from Mozilla Common Voice [10], release `cv-corpus-26.0-2026-06-12`, Hindi (`hi`) locale (11,131 rows in `validated.tsv`). Candidates are filtered on: (i) community vote margin (`up_votes`−`down_votes`) ≥ 2 (ii) duration in $[2.0, 8.0]$ s (iii) a per-speaker cap of 5 clips in order to prevent a small number of speakers from dominating the set.

The vote margin figure was chosen empirically, a full-corpus sweep showed `Min_vote_margin=1` is not viable on this release since all 11,131 rows already satisfy it. `Min_vote_margin=3` on the other hand collapses the pool to 225 rows. `Min_vote_margin=2` is therefore chosen as the threshold since it removes a substantial fraction of borderline clips (9,863 of 11,131 rows pass) without collapsing the pool. The full corpus candidate scan is summarized in Table 7.

This paper targets **1,125 pairs** (ceiling minus a $\approx$5% safety margin for synthesis time failures), with `oversample_factor=1.05`.

The mel$\rightarrow$linear `InverseMelScale` step uses SGD pseudo inversion.

Table 7: Full corpus candidate scan at `Min_vote_margin=2`, `max_clips_per_speaker=5`, duration $\in [2.0, 8.0]$s.

Stage	Count
Total rows scanned	11,131
Failed vote-margin filter	1,268
Hit per-speaker cap	8,583
Missing audio file	0
Failed duration window	95
True achievable candidates	1,185
Unique speakers used	340

Table 8: `InverseMelScale` SGD iteration sweep, mean log mel L1 distance (dB) vs. independent librosa reconstruction. Confirmed at $n = 10$ and $n = 30$ samples up to 4,000 iterations.

`inv_mel_max_iter`	Mean log mel L1 (dB)
1,000	3.64–4.00
2,000	2.17–2.23
2,250	2.10
2,500	2.04–2.09
3,000	2.01–2.02
3,500	1.98–1.99
4,000	2.00

`inv_mel_max_iter` is set to 5,000 (with SGD momentum 0.99 and learning rate 0.1) which is comfortably past the documented plateau region, as a conservative choice (Table 8). The residual $\approx$2 dB gap at plateau highlights the expected difference between two distinct pseudo inverse algorithms (SGD vs. librosa's NNLS based approach) instead of reflecting an unconverged optimization.

Table 9: Realized Hindi Griffin-Lim dataset.

Pairs produced	1,125 / 1,125 target
Unique speakers (selection / realized)	340 / 338
Mean / min / max duration (s)	5.14 / 2.02 / 7.99
RMS matched pairs	1,081 (96.1%)
Peak-limited pairs (0.99 fallback)	44 (3.9%)
Mean relative RMS error, RMS-matched pairs	0.11%
Mean spectral L1 (real vs. fake, dB)	25.67 ± 8.89 (median 27.88)

Realized Dataset All 1,125 targeted pairs were successfully produced. Quality control on the finalized metadata confirms that the synthesis procedure achieved its intended amplitude and duration matching with the peak-limited fallback being applied to a small subset as shown in Table 9, clipping is inherited from the source recordings instead of being introduced by the synthesis procedure.

Despite intended amplitude and duration matching, the mean real vs fake spectral L1 distance quantifies the distinction between Griffin Lim reconstructions and their source recordings and is the spectral gap the detector is expected to attend to.

3 Experimental Setup

3.1 Detection Model

Pretrained wav2vec 2.0 XLS-R (300M) + AASIST checkpoint from `TakHemlata/SSL_Anti-spoofing` [1], has been used in this paper without additional fine tuning. The model uses a 4.0375 s and 16 kHz waveform to predict bonafide/spoof logits as outputs. The bonafide logit is used as the detection score throughout.

3.2 System Configuration

Table 10: System Configuration

Component	Specification
Python	3.8.20
torch	1.13.1
torchaudio	0.13.1
GPU	NVIDIA DGX A100, 40 GB VRAM

3.3 Datasets

English – ASVspoof 2019 LA eval. contains 71,237 utterances (7,355 bonafide, 63,882 spoofed) which span across 13 attack types (A07–A19) generated with different vocoders [8].

Common voice Griffin Lim datasets. Controlled construction of 1,125 paired utterances of real/fake audio using identical Griffin Lim copy synthesis across both languages.

3.4 Explainability Methods

Occlusion. Importance of each sample is calculated with overlapping windows (20/50/100 ms, 50% step overlap), The bonafide logit score is compared with and without that sample for each window. Mean absolute score delta across all windows covering that sample is its importance.

Integrated Gradients (IG). Captum's IG implementation [13] is used starting with a zero baseline. 50 interpolation steps are made successively, and for each sample the change in the bonafide logit score is observed at each of these interpolation steps.

3.5 Faithfulness Metrics

Attribution maps use **comprehensiveness**: defined by the drop in bonafide logit score when top-K attributed samples are masked to zero, and **sufficiency**: defined by the bonafide logit score when only top-K attributed samples are kept, with everything else masked to zero.

Higher comprehensiveness and sufficiency = more faithfulness.

3.6 Silence / Shortcut Analysis

silence detection is done with `librosa.effects.trim/split` at a threshold of `top_db` = **40**. The regions of silence are identified prior to any padding or feature extraction.

The effect of trimming silence, prediction based on silence duration, and attribution overlap with detected silence is studied.

3.7 Feature Level Explanation Drift Metrics

Multiple feature level metrics are computed to analyze the explanation drift across languages. STFT weighting is used to partition the signal into low (0–1 kHz), mid (1–4 kHz), and high (4–8 kHz) bands serving as an energy weighted frequency attribution proxy. Attribution placed on speech vs silence regions helps in their comparison. Gini index and Shannon entropy provide a measure of temporal concentration.

3.8 Robustness Controls

To account for acoustic differences, overlap weighted EER analysis and spectral caliper matching condition on fake audio low/mid/high energy fractions, RMS, peak amplitude, and duration prove useful and caliper analysis changes the target population.

4 Results

In this section, we focus on evaluating whether the detector's in domain English behaviour extends to Hindi Griffin Lim copy synthesis and whether the input level explanations differ.

Matched synthesis, preprocessing and dataset construction protocol allows for a Hindi Common Voice Griffin Lim vs English Common Voice Griffin Lim comparison to test a language associated degradation.

Table 11: Detection performance across benchmark and Common Voice conditions. CIs use 2,000 speaker-cluster bootstrap resamples.

Condition	EER (%)	95% CI
ASVspoof 2019 LA eval (all attacks)	0.22	–
English CV Griffin–Lim control	16.18	[14.34, 17.56]
Hindi CV Griffin–Lim copy-synthesis	30.49	[27.67, 33.25]
Hindi CV, RMS-matched subset (sensitivity)	30.34	[27.51, 33.09]

4.1 Detection Performance

The matched English CV condition control has a substantially higher EER than the ASVspoof benchmark (16.18% EER, Table 11). The EER further increases in case of Hindi (30.49% EER), a 14.31 point gap (95% CI [11.38, 17.92]). It can be observed that the result is not driven by the 44 peak-limited Hindi reconstructions since restricting Hindi to the 1,081 RMS-matched pairs changes EER by only −0.15 points. Figure 7 shows the primary comparison.

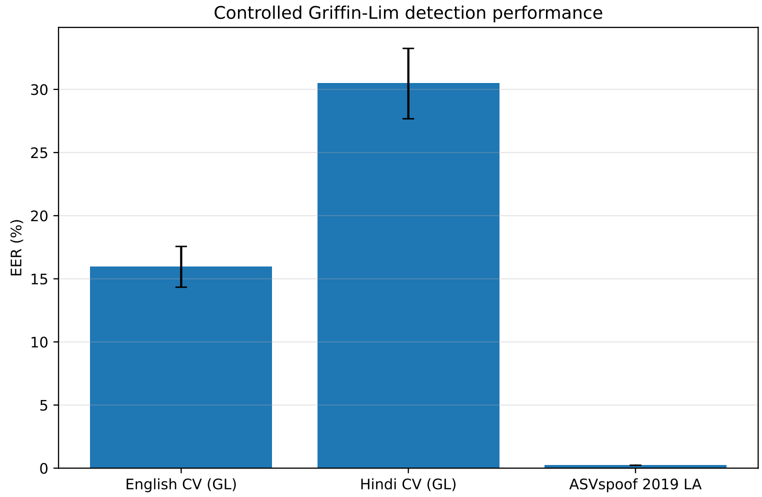

Fig. 7: Controlled Griffin Lim detection performance. Error bars are 95% speaker-cluster bootstrap CIs. ASVspoof 2019 LA is an in-domain benchmark, the English/Hindi Common Voice comparison is the primary matched copy synthesis condition.

Table 12 shows that the gap survives controls for measured non linguistic differences. Overlap weighting keeping all pairs but focusing on their common acoustic support, leaves an 8.52 point gap. Spectral caliper matching shows gaps of 9.21–12.10 points across three strictness levels. Further, a 12.80 point gap is observed after trimming leading/trailing silence independently from every utterance.

Table 12: Robustness of the Hindi minus English EER gap. Acoustic controls use low/mid/high energy fractions, RMS, peak, and duration. CIs are 95% speaker cluster bootstrap intervals.

Analysis	Gap (pp)	95% CI
Primary, all pairs	14.31	[11.38, 17.92]
Acoustic-overlap weighting	8.52	[4.56, 12.76]
Spectral caliper, 1.0 SD (n=248 pairs)	12.10	[7.40, 17.59]
Spectral caliper, 1.5 SD (n=471 pairs)	9.55	[5.86, 14.23]
Spectral caliper, 2.0 SD (n=717 pairs)	9.21	[5.24, 13.14]
Silence-trimmed waveforms	12.80	[9.54, 16.12]

The result is also visible at the speaker level. For each speaker, we compute the mean real bonafide logit minus the mean fake bonafide logit (larger values indicate cleaner real/fake separation). The analysis shows that Hindi speakers have a smaller mean separation margin (2.30 logits) than English speakers (5.72 logits): mean Hindi minus English difference −3.42, 95% CI [−3.88, −2.98]. Moreover, 25.4% of Hindi speakers have a non positive separation margin, compared with 4.9% of English speakers (Table 13). This distributional heterogeneity further supports the EER result.

4.2 Explanation Analysis and Faithfulness

The same occlusion and Integrated Gradients pipeline used earlier is applied to 100 real/fake pairs per Common Voice language, and a secondary reference of 100 sampled ASVspoof A11 attacks and 100 matched bonafide utterances.

Table 13: Speaker level real/fake separation. A speaker's margin is defined as the difference between their mean real bonafide logit and their mean fake bonafide logit, non positive values indicate that the average real bonafide logit is not higher than its fake counterpart.

Metric	English (n=974)	Hindi (n=338)
Mean separation margin	5.72	2.30
Median separation margin	6.05	2.85
SD of separation margin	3.34	3.79
Speakers with non positive margin	4.9%	25.4%

Table 14: IG faithfulness at top K=10% (mean ± std). All attributions target the bonafide logit; rows are reported separately by true class because their score distributions differ.

Condition	Class	Comprehensiveness	Sufficiency
English CV	real	6.22 ± 2.40	-3.95 ± 2.21
English CV	fake	1.33 ± 2.44	-5.75 ± 1.47
Hindi CV	real	6.17 ± 2.44	-4.03 ± 2.02
Hindi CV	fake	0.15 ± 2.62	-4.71 ± 2.57

For real utterances in both languages, masking the most-attributed 10% of samples significantly reduces the bonafide logit (≈ 6.2 logits, Table 14), interestingly keeping only those samples is not sufficient.

For fake utterances, absolute attribution values should not be seen as a direct measure of detector confidence since comprehensiveness values are lower and more distinct.

Hence, we focus on the shift in attribution features between real-fake pairs, instead of relying on faithfulness alone, to diagnose explanation drift.

4.3 Feature Level Explanation Drift

In order to diagnose cross lingual explanation drift, we compare the primary attribution features between matched English and Hindi CV conditions.

We follow that by analyzing paired real–fake changes to differentiate between a reconstruction associated shift and a baseline difference between the language datasets.

Energy-weighted frequency band measures are used as descriptive proxies for where the input attribution is concentrated, instead of evidence that a specific frequency band is inherently more important.

Table 15: Energy-weighted frequency attribution proxy for fake utterances. Δ is Hindi minus English, CIs are speaker-cluster bootstrap intervals and q values use Benjamini–Hochberg correction.

Band	English (%)	Hindi (%)	Δ pp (95% CI)	q
Low (0–1 kHz)	73.48	44.58	−28.90 [−33.40, −24.27]	$< 10^{-19}$
Mid (1–4 kHz)	19.82	30.81	+10.99 [+8.35, +13.56]	$< 10^{-13}$
High (4–8 kHz)	6.70	24.61	+17.91 [+15.78, +19.91]	$< 10^{-22}$

Figure 8 and Table 15 note a large redistribution of fake attribution toward mid/high-frequency energy. Baseline cross language acoustics could be reflected by this comparison itself. It can be observed that the paired interaction is stronger – reconstruction changes low/mid/high attribution by $-12.01/+8.32/+3.69$ points in English, but by $-45.03/+22.07/+22.95$ points in Hindi. Thus the resulting Hindi minus English reconstruction interaction is −33.02 points for low frequency (95% CI [−36.70, −29.28]), +13.75 for mid frequency ([11.71, 15.61]), and +19.27 for high frequency ([17.16, 21.27]). The same sign pattern with nonzero confidence intervals is observed with 20, 50, and 100 ms occlusion maps, indicating that the result is not specific to IG.

Table 16: Speech vs. Silence attribution and temporal concentration (fake condition).

Metric	English CV	Hindi CV
Speech attribution fraction	0.99996	0.99840
Silence attribution fraction	0.00004	0.00160
Gini coefficient	0.731	0.721
Shannon entropy	14.374	14.400
Top 1% concentration	0.164	0.174
Top 5% concentration	0.422	0.421
Top 10% concentration	0.587	0.578

The speech/silence difference is insignificant after Benjamini–Hochberg correction (q=0.078). Gini (q=0.339) and Shannon entropy (q=0.910) also do not differ detectably. Meanwhile, Top 1% concentration

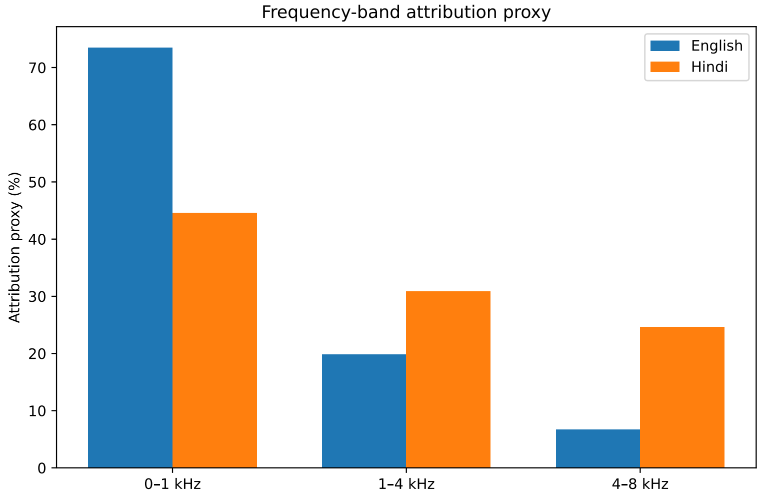

Fig. 8: Energy-weighted frequency attribution for Common Voice fake utterances. The figure summarizes the primary fake condition comparison, Inferential statistics are given in Table 15.

increases by only 1.00 point (q=0.029) and the top 5% and 10% concentrations are essentially unchanged after correction (Table 16). Therefore, the dominant shift occurs within active speech particularly across frequency bands, and not marked by a shift toward silence or a broad change in temporal sparsity.

4.4 Silence Analysis on Hindi CV and English CV

The silence analysis in Section 2.3 is repeated on the raw English CV and Hindi CV recordings before model padding. Duration-specific separation is reported and EER is recomputed after applying the same deterministic boundary-silence trim to every real and fake utterance.

Table 17: Silence analysis on the Common Voice conditions (n=1,125 paired real/fake utterances per language). Wilcoxon tests are paired real versus fake silence duration comparisons.

Quantity	English CV	Hindi CV
Leading silence only EER (%)	14.49	10.49
Trailing silence only EER (%)	20.62	17.33
Mean leading silence, real / fake (ms)	410 / 0	470 / 44
Mean trailing silence, real / fake (ms)	445 / 0	573 / 58
Wilcoxon p, leading silence	3.66×10^{-159}	2.73×10^{-154}
Wilcoxon p, trailing silence	9.35×10^{-148}	7.39×10^{-151}
EER after silence trimming (%)	43.11	55.91

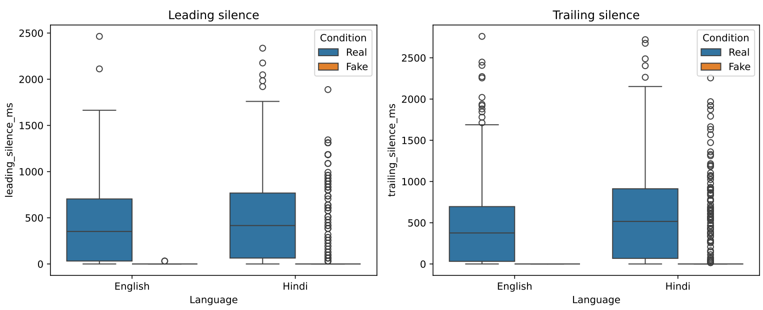

Fig. 9: Leading and trailing silence distributions on the matched CV conditions. The artifact is present in both languages, motivating the trimmed waveform robustness test reported in Table 12.

Both CV conditions contain a strong real/fake boundary-silence artifact, reflected by the paired duration differences being overwhelmingly significant ($p < 10^{-147}$), as reported in Table 17 and Figure 9. The EERs corresponding to the silence duration alone substantially differentiates real and fake examples. However, it still cannot explain the result between languages: Hindi remains 12.80 points worse than English even after removing boundary silence from every real and fake clip (Table 12). This differs in context from our earlier silence shortcut analysis in which silence trimming caused a major increase in EER within ASVspoof, and supports the conclusion that the explanation drift currently being studied is not simply a reappearance of the silence shortcut.

5 Discussion

The results provide a more specific explanation than the EER gap alone. Persistence of the Hindi–English EER gap even after silence trimming and acoustic controls shows that boundary silence cannot fully explain the performance degradation observed in the Hindi condition (Table 12). However, silence is still relevant in both conditions marked by significant difference in its durations in real and fake clips. Despite attribution being mainly placed on speech, the ASVspoof shortcut analysis shows that silence can still affect detector decisions. Therefore, the conclusion is not limited to the low attribution found in silence, it is also based on the EER gap that remains after silence trimming.

In case of fake utterances, the major attribution change is a shift from the low frequency band towards the mid/high frequency bands in the Hindi condition (Table 15, Figure 8). Paired real fake comparison shows that the reconstruction related change is much larger in Hindi as compared to English. The same pattern is observed with Integrated Gradients and all three occlusion windows. In contrast, attribution remains mainly on active speech, while broad temporal concentration is similar across the two conditions (Table 16) suggesting that the main change in explanations occurs within speech and is not a general shift towards silence. It must be noted that the frequency band measures are attribution proxies and do not by themselves identify a causal frequency artifact.

6 Comparative Analysis with Existing Work

Table 18

Our Work	**Previous Work**
This paper uses the same pretrained wav2vec 2.0 XLS-R + AASIST ($\approx$ 318M parameters) checkpoint from [1] without any additional training or fine tuning. The study uses this model to analyze the performance gap in cross-lingual deepfake detection using post-hoc explainability.	**SSL based deepfake detection.** Tak et al. [1] report state of the art results on ASV spoof 2021 LA and DF, fine tuning a wav2vec 2.0 XLS-R (300M) frontend and coupling it to the AASIST graph attention backend [2].
This study replicates the occlusion/IG and the faithfulness analysis inspired by [3] on English. The analysis is then extended to a controlled English–Hindi CV comparison.	**xAI for audio deepfakes.** Grinberg et al. study explainability methods analysing the specific time domain regions that an audio deepfake detector attends to [3]. They propose a diffusion based explanation method [4].
In this paper, a different SSL based detector is used to reproduce the silence shortcut established in [5]. The shortcut is observed to be predictive but does not account for the cross lingual performance gap.	**Shortcut learning in spoofing detection.** Müller et al. note that the bonafide/spoof label on ASV spoof can be predicted with 85% accuracy / 15.1% EER by leading silence duration alone. They find performance degradation in RawNet2 based detector after trimming silence [5].
Instead of proposing a new training intervention, this study discovers a shift in frequency band attribution proxy: forming one of the explanations for the gap in performance.	**Related directions.** Teissier et al. show that deepfake detection generalization is improved by encouraging sparse, disentangled representations [6].
Instead of layer probing, this study probes input level attribution through occlusion and IG. A possible future direction is combining input and layer level analysis.	**Related directions.** Beheshti et al. layer probe the SSL model to find which layers carry important spoof relevant information across domains [7].

7 Limitations

- **Single model.** All results are specific to a wav2vec 2.0 XLS-R plus AASIST checkpoint and may not transfer to other architectures or training recipes.
- **Language and synthesis scope.** We evaluate one non English language and one Griffin Lim copy synthesis pipeline. The observed pattern may not generalize to other languages, parallel multilingual speech, or independent vocoder and neural attacks.
- **Common support precision.** The overlap weighted estimate targets specific regions (acoustic common support) instead of the full corpora. Effective sample sizes are brought down to roughly 302 English and 342 Hindi pairs.

8 Conclusion and Future Work

This study uses post-hoc explainability to understand the performance gap between the detection of English and Hindi audio deepfakes. Under the matched Common Voice Griffin Lim copy synthesis protocol, the wav2vec 2.0 XLS-R + AASIST detector has an EER of 16.18% on English and 30.49% on Hindi, a 14.31 point difference (95% CI [11.38, 17.92]). The gap persists even after boundary silence trimming, acoustic support weighting and spectral-caliper matching. For fake utterances, this gap is accompanied by a shift in energy weighted frequency band attribution within active speech. However, these results do not establish a causal effect of Hindi or any particular frequency band on the performance degradation.

Future work should repeat this analysis using parallel multilingual speech with the same speakers, text and similar recording conditions. The analysis should also be extended to more detector architectures and synthesis families. Further, controlled filtering, masking, or resynthesis of the frequency regions can identify whether they directly cause a change in detector scores and help identify useful multilingual training or augmentation strategies. Another possible direction is to study the internal representations of the model using techniques like layer probing and activation patching. This can show whether different layers use different spoof related features for English and Hindi, and whether the internal representations meaningfully contribute to the observed performance gap.

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